

Homework for Introduction to Mathematics

1. Solve the following questions with the specified method.

- (a) Negate $(P \Rightarrow Q) \Rightarrow P$, you don't need to simplify it.
- (b) Using $((P \wedge Q) \vee P) \equiv P$ to justify $\sim [(P \Rightarrow Q) \Rightarrow P]$ and $\sim P$ are logically equivalent.
- (c) State the definition of logically equivalent, and using it to justify

$$((P \Rightarrow Q) \Rightarrow P) \Leftrightarrow P$$

is tautology.

Proof.

(a)

$$\begin{aligned}\sim ((P \Rightarrow Q) \Rightarrow P) &\equiv \sim (\sim (P \Rightarrow Q) \vee P) && (P \Rightarrow Q \equiv (\sim P) \vee Q) \\ &\equiv (P \Rightarrow Q) \wedge (\sim P) && \text{(De-Morgan's Law)} \\ &\equiv ((\sim P) \vee Q) \wedge (\sim P) && (P \Rightarrow Q \equiv (\sim P) \vee Q)\end{aligned}$$

- (b) We want to show $\sim [(P \Rightarrow Q) \Rightarrow P] \equiv \sim P$, simplifying $(P \Rightarrow Q) \Rightarrow P$, we have

$$\begin{aligned}(P \Rightarrow Q) \Rightarrow P &\equiv \sim (P \Rightarrow Q) \vee P \\ &\equiv (\sim (\sim P \vee Q)) \vee P \\ &\equiv (P \wedge (\sim Q)) \vee P\end{aligned}$$

Using $(P \wedge Q) \vee P \equiv P$, we obtain

$$(P \wedge (\sim Q)) \vee P \equiv P,$$

that is

$$(P \Rightarrow Q) \Rightarrow P \equiv P.$$

Hence

$$\sim [(P \Rightarrow Q) \Rightarrow P] \equiv \sim P.$$

- (c) S_1 and S_2 are logically equivalent if and only if $S_1 \Leftrightarrow S_2$ is a tautology. By (b), we know $(P \Rightarrow Q) \Rightarrow P \equiv P$, so hence $(P \Rightarrow Q) \Rightarrow P \Leftrightarrow P$ is tautology.

□

2. For the following statement forms, write down their logically equivalent statement forms. **Only using $\sim$ and $\vee$.**

(a) $P \Leftrightarrow (\sim Q)$.

(b) $(P \vee Q) \Rightarrow (P \wedge Q)$.

(c) $(P \Rightarrow Q) \vee (Q \Rightarrow P)$.

Sol.

(a)

$$\begin{aligned} P \Leftrightarrow (\sim Q) &\equiv (P \Rightarrow (\sim Q)) \wedge (\sim Q \Rightarrow P) \\ &\equiv ((\sim P) \vee (\sim Q)) \wedge (Q \vee P) \\ &\equiv \sim [\sim ((\sim P) \vee (\sim Q)) \vee \sim (Q \vee P)] \\ &\text{(or } \equiv \sim [\sim ((\sim P) \vee (\sim Q)) \vee \sim (P \vee Q)]) \end{aligned}$$

(b)

$$\begin{aligned} (P \vee Q) \Rightarrow (P \wedge Q) &\equiv \sim (P \vee Q) \vee (P \wedge Q) \\ &\equiv \sim (P \vee Q) \vee \sim ((\sim P) \vee (\sim Q)) \end{aligned}$$

(c)

$$(P \Rightarrow Q) \vee (Q \Rightarrow P) \equiv ((\sim P) \vee Q) \vee ((\sim Q) \vee P)$$

3. Let P be the statement: If $x^2 - 3x + 2 < 0$, then $1.3 < x < 1.7$. Write $\sim P$.

Sol. The statement P can be formally written as:

$$\forall x \in \mathbb{R}, (x^2 - 3x + 2 < 0) \Rightarrow (1.3 < x < 1.7)$$

Since $\sim (A \Rightarrow B) \equiv A \wedge (\sim B)$, the negation of P is:

$\sim P$: There exists $x \in \mathbb{R}$ such that $x^2 - 3x + 2 < 0$ and $(x \leq 1.3 \text{ or } x \geq 1.7)$

(Alternatively: There exists $x \in \mathbb{R}$ such that $x^2 - 3x + 2 < 0$ and $x \notin (1.3, 1.7)$.)